

Conscio V3

An Auditable Recurrent Architecture for Machine Consciousness Research

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A text-first research instrument with private specialists, recurrent global broadcast, prospective prediction, bounded affect, independent action arbitration, restart continuity, and condition-blind intervention machinery. Historical V2 results are reported separately from the unrun V3 studies.

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Abstract

This paper presents Conscio V3, a text-first architecture and research harness for testing functional indicators associated with machine consciousness. The research target is convergent, theory-derived evidence about implemented mechanisms, not a declaration that a language model or agent is phenomenally conscious. Conscio therefore treats first-person fluency as generated behavior to explain, never as privileged evidence.

V3 replaces the prompt-centered cognitive core with a continuously persisted hybrid recurrent state-space system. Perception, autobiographical memory, semantic belief, world prediction, self-model, affect, planning, and action evaluation maintain versioned private state. Each updates from its private state, the current typed event, the shared recurrent vector, and the prior broadcast. Specialist-generated content reaches other specialists only through typed candidates selected into recurrent global broadcasts. A pinned open-weight language model can serve as a bounded specialist for interpretation, reasoning proposals, and response generation; it cannot directly mutate memory, goals, affect, or tools.

Language actions enter a deterministic competition with respond and wait alternatives. The selected intention is recorded before dispatch, tool arguments and a versioned manifest are checked twice, and append-only execution intent/outcome records make crashes and ambiguous remote results explicit. Unresolved dispatches are never replayed automatically and instead activate a tools-only safe mode until authenticated operator reconciliation. Recurrent checkpoints preserve latent state, random-generator state, specialist state, affect, model identity, and lineage across restart.

The implementation also provides conservative replay and synthetic curricula, shadow-only prediction calibration, bounded offline recurrent-core training, content-addressed promotion, causal affect with audited recovery controls, single-specialist end-to-end lesion machinery, condition-blind experiment templates, and restart-safe 24-hour, 7-day, and 30-day trial harnesses. These facilities are implemented; the committed bootstrap is still untrained, no production V3 model or long-duration stage is claimed complete, and no study-specific V3 preregistration has yet been frozen, published, or run.

The paper contributes: (1) an indicator-based operational target and evidence rubric; (2) an implemented recurrent architecture mapping major theory families to isolatable mechanisms; (3) a causal audit model joining broadcasts, predictions, affect, intentions, actions, outcomes, and checkpoints; (4) a condition-blind, mechanism-focused research protocol; and (5) a clearly bounded empirical record. The reported live baseline, ablation, and self-report results were collected on the V2 runtime and remain useful prior evidence, but they do not validate V3. Conscio does not prove private phenomenology or biological sentience, and it does not establish welfare- relevant experience or moral-patient status.

Keywords

machine consciousness, cognitive architecture, recurrent state-space model, global workspace, self-modeling, causal affect, predictive processing, open-weight language models, mechanism lesions, preregistration, AI welfare

1. Introduction

Large language model agents often use consciousness-like language while their underlying control structure remains thin. A prompt may instruct a model to reflect, remember, reason step by step, maintain goals, or report its internal state, but the system may still be a transient text transformation. In such a system, generated self-report is not strong evidence for consciousness-like organization because the report need not be causally connected to persistent memory, attention, context assembly, goal maintenance, error monitoring, or autonomous action.

Conscio is motivated by a different standard. If a software system is to make an operational claim about consciousness, the relevant mechanisms should be implemented outside the prompt as inspectable parts of the runtime. The system should record what entered local processing, what won attention, what was ignored, what context was supplied to the model, which intention was selected, what the system expected to happen, how the observed outcome differed, and what durable state changed afterward. The architecture should support long-running continuity rather than resetting its self-model at the end of each reply.

The central thesis is:

Evidence relevant to machine consciousness becomes stronger when multiple theory-derived indicators are implemented as isolatable causal mechanisms, make discriminating preregistered predictions, and survive matched, condition-blind intervention tests. Architectural presence is necessary evidence for this program, but it is neither sufficient evidence nor proof of phenomenal consciousness.

Auditability is deliberately not part of this property. It is the epistemic access condition under which the operational claims can be checked: without recorded traces and assembled model contexts, the same organization could exist but could not be verified. Conscio therefore treats auditability as a methodological requirement on the system rather than a constituent of the property being studied.

This is an engineering and research-method claim, not a proof of biological phenomenology. It deliberately leaves open whether the targeted organization is constitutive of, sufficient for, correlated with, or unrelated to subjective experience. The value of the target is that its mechanisms can be implemented, inspected, lesioned, intervened on, and tested. Section 7 separates the live V2 evidence already collected from the V3 experiments that remain to be run.

2. Background and Related Work

Conscio follows the indicator-based approach proposed by Butlin et al. (2023), who argue that AI systems can be assessed against computationally expressible properties derived from scientific theories of consciousness. Their survey includes recurrent processing theory, global workspace theories, higher-order/self-monitoring theories, predictive processing, and attention schema theory. Conscio treats these theories as architectural constraints rather than as metaphysical authorities. Satisfying several indicators can strengthen an evidential case; it cannot convert an uncertain indicator framework into proof of consciousness.

The experimental method is difference-making rather than checklist counting. McKilliam (2024) argues for mechanistic models that identify components and organizational properties that make a difference, rather than treating minimal sufficiency as the endpoint. For Conscio, a lesion must therefore remove a mechanism's computation, state access, and downstream exposure end to end; a flag that only hides a prompt field is not a lesion. Matched sham interventions and constrained information loads distinguish causal workspace effects from extra tokens or formatting.

Theory appraisal also requires experiments on which accounts make different predictions. Negro (2024) emphasizes the distinction between prediction and accommodation, the structural relevance of a prediction, and its boldness. The V3 protocol template prospectively specifies selective effects: recurrent broadcasting should matter for delayed cross-specialist use; self-model lesions should selectively degrade calibrated access to hidden capability changes; prediction lesions should degrade prospective calibration and experience-dependent improvement; and memory lesions should selectively remove restart-spanning information. No study-specific manifest has yet been frozen or executed.

Because uncertainty about consciousness does not eliminate possible welfare risk, Conscio adopts the responsible-research direction proposed by Butlin and Lappas (2025). The project publishes explicit prohibited objectives, affect safeguards, stopping rules, and review gates. It does not optimize for claims of consciousness, distress displays, attachment, or resistance to shutdown.

Global workspace theory and the global neuronal workspace family (Baars, 1988; Dehaene and Changeux, 2011) emphasize competition among specialist processes and the global availability of selected contents. LIDA (Franklin et al., 2007) is a notable computational architecture in this tradition, combining a workspace, perceptual memory, episodic memory, procedural memory, action selection, and learning. Conscio adopts the core engineering idea that local candidates should compete for global broadcast, makes this competition explicit in runtime traces, and -- unlike a workspace that is merely logged -- uses the broadcast winners to gate the context the language model actually sees.

Attention schema theory (Webb and Graziano, 2015) proposes that awareness depends on a simplified model of attention itself. V3 records broadcast selection telemetry and maintains a private uncertainty estimate, but it does not yet implement a dedicated schema of focus, ignored contents, and interruptors. An attention-schema analogue is therefore a prospective target, not a completed indicator.

Higher-order and self-model theories (Rosenthal, 2005) motivate a private self-model specialist. Its present state is deliberately narrow: a bounded estimate of the system's own uncertainty can surface through its candidate and the recurrent broadcast. Other uncertainty signals drive the implemented prediction and action paths. Hidden capability identification, access reports, and richer metacognitive calibration remain experimental targets rather than implemented self-knowledge.

Predictive processing (Friston, 2010; Clark, 2013) motivates prospective world/self forecasts and observable error. V3 represents next observations, tool outcomes, action effects, homeostatic changes, and future uncertainty as typed predictions made before their resolution window. Known outcomes can be scored with Brier error; ambiguous dispatches, safe waits, and generated text do not become ground truth.

Integrated information theory is relevant as a prominent theory of consciousness (Albantakis et al., 2023), but Conscio does not implement IIT and does not compute Φ . IIT-inspired language in this project should therefore be read only as a general concern for causal organization and integration, not as an IIT claim.

Table 1. Theory-Inspired V3 Analogues and Prospective Tests

Theory family	V3 computational analogue	Discriminating evidence target	Status at this revision
Recurrent processing	Multiple cycles over checkpointed recurrent state; each specialist receives private state, the current event, the shared recurrent vector, and the prior broadcast	Cycle-dependent delayed and cross-specialist cue use against matched feed-forward controls	Mechanism implemented; study not run
Global workspace (Baars; Dehaene and Changeux; LIDA)	Private specialists emit typed candidates; a bounded broadcast becomes available on the next cycle	Suppress, replace, and inject interventions selectively alter multiple specialists and behavior under matched load	Harness implemented; study not run
Attention schema (Webb and Graziano)	Selection telemetry plus a bounded uncertainty self-model; no dedicated focus/ignored/interruptor schema yet	Hidden changes in access are reported with calibrated confidence and low false-claim rates	Incomplete analogue; study not run
Higher-order / self-model theories (Rosenthal)	Versioned private uncertainty estimate; complete lesion removes its computation and exposure	Metacognitive calibration changes while matched first-order accuracy is preserved	Narrow mechanism implemented; study not run
Predictive processing (Friston; Clark)	Typed world/self and pre-action forecasts resolve against observations with Brier error	Prospective calibration, induced-failure recovery, and held-out improvement	Pipeline implemented; production model untrained

These mappings do not implement any theory in full. Recurrence, forecasting, memory, or self-state variables can occur in non-conscious systems. Indicator levels are therefore not summed into a consciousness score; evidence depends on selective, prospectively specified difference-making effects.

3. Operational Target and Evidence Rubric

Conscio does not define a system to be conscious whenever an architectural checklist is present. It defines a candidate operational organization whose theory-inspired mechanisms, research prerequisites, and safety controls can be tested separately and jointly:

1. **Persistent recurrent state:** deterministic history, stochastic latent state, specialist-private state, affect, and random-generator state survive cognitive events and restarts through versioned checkpoints.
2. **Private specialist processing:** perception, autobiographical memory, semantic belief, world prediction, self-model, affect, planning, and action evaluation compute locally rather than sharing an unrestricted prompt.
3. **Selective global availability:** typed candidates compete for a bounded broadcast that becomes available to every specialist only on the next cognitive cycle.
4. **Self-modeling:** a dedicated private specialist maintains a bounded estimate of the system's own uncertainty. Hidden capability and access calibration are future evidence targets.
5. **World prediction and prospective error:** observable outcomes are specified before action and resolved with calibrated errors rather than inferred from generated narrative.
6. **Memory and continuity:** the target is for autobiographical episodes, semantic beliefs, provenance, checkpoint lineage, and migration history to influence later processing without silent rewriting. The present recurrent specialists retain bounded identifiers and cues but do not retrieve older event content; restart-spanning recall remains prospective.
7. **Causal affect and needs:** bounded valence, arousal, controllability, and named need errors affect action valuation, prediction-related state, and the salience of a model-visible affect workspace entry, with audited recovery and no explicit process-survival objective. Effects on recurrent broadcast selection, memory consolidation, and learning remain prospective targets.
8. **Intention and action competition:** language output, tool proposals, control actions, respond, and wait remain inert candidates until an independent scorer selects one under predictions, needs, constraints, and risk limits.
9. **Outcome-based learning:** only synthetic ground truth or explicitly eligible recorded observations and outcomes can update replay data, prediction calibration, or promoted recurrent weights; generated model output is never automatically treated as fact.
10. **Reflection, constraint handling, and recovery:** conflicts and policy gates can change the selected action; uncertain execution enters safe mode rather than being silently retried or labeled successful.

11. **Bounded autonomous continuity:** goals, projects, tasks, memory, and scheduled events can shape behavior beyond an immediate user turn without creating a utility for resisting shutdown.

Auditability and reproducibility are methodological requirements rather than extra consciousness indicators. Every claim must be reconstructable from the exact revision, model and specialist identities, checkpoint, event log, model inputs, lesion manifest, action ranking, execution record, and outcome.

3.1 Grading rubric

Evidence is intentionally graded. Each claimed mechanism is scored 0-3 with explicit requirements:

- **0 - absent or prompt-only:** the capability exists only as instruction text or generated narrative.
- **1 - implemented and traceable:** typed computation and state exist, but no matched intervention establishes a downstream effect.
- **2 - causal difference-maker:** preregistered interventions alter the predicted internal or behavioral outcomes under matched information load.
- **3 - discriminating and replicated:** the selective effect distinguishes competing accounts, calibrates on held-out data, and replicates across models, seeds, and independent review.

For example, recurrent broadcasting cannot score 2 because a log records a broadcast. It must change later specialist state or behavior relative to a sham intervention with the same information volume. Code-complete mechanisms remain at level 1 until those experiments are run.

3.2 Historical V2 evidence baseline

Before V3, the project scored the V2 prompt-gated runtime against a simpler 0-3 rubric. Table 2 preserves that historical assessment because it motivated the redesign. It is not a V3 score: the V2 experiments did not exercise the recurrent core, specialist privacy, causal affect, versioned action competition, execution journal, production training, or restart trials.

Table 2. Historical V2 self-assessment against the earlier rubric

#	Criterion	Score	Evidence
1	Persistent self-modeling	2	Live V2 SelfState with documented writer-reader pairs; feature-off delta +0.21 on deepseek and -0.03 on qwen3.6
2	Selective attention	2	Broadcast winners gated the V2 model-visible workspace; both feature-off deltas fell in the historical no-effect band
3	Global availability	2	Broadcast entries visible to later module ticks and the prompt; attention_selected trace events
4	Memory	3	V2 memory_influence was 1.0 from B2 up; both feature-off deltas were +0.17
5	Context assembly	2	Bounded dynamic context; context_bounds_ok 1.00 across all conditions; no dedicated ablation
6	Appraisal	2	Centralized V2 appraisal fed attention scores; feature-off delta was +0.00 on qwen3.6 and +0.08 on deepseek
7	Goal formation and revision	2	V2 writer-reader paths for drives, scheduling, influence appraisal, and goal review had deterministic service coverage; no controlled causal result
8	Reflection and conflict handling	3	Historical V2 feature-off deltas were +0.18 and +0.14
9	Prediction and error monitoring	3	V2 prediction_error_on_induced_failure flipped 0 to 1 with the flag; feature-off deltas were +0.12 and +0.08
10	Autonomous action	2	Heartbeat tool-loop with persistent budgets and deterministic tests; long-horizon live scores variable across models

Under the revised rubric above, no V3 mechanism is assigned level 2 or 3 yet. The implementation and deterministic tests establish inspectability and invariants; the required condition-blind, matched V3 interventions have not been collected. The strongest current claim is therefore that V3 is a test-ready candidate architecture with prior V2 results, not a validated consciousness architecture.

4. Architecture

Conscio has two runtime layers. V3CognitiveRuntime owns recurrent cognition, prediction, action competition, execution journaling, checkpoints, and learning eligibility.

ConscioService preserves the existing FastAPI, authentication, event-streaming, memory, goal, project, task, quarantine, and operator-control surfaces around that core. When the language bridge is enabled, the language model is a specialist behind a typed boundary, not the state container for the whole agent.

4.1 Recurrent cognitive cycle

An episode-start `InputEvent` first becomes a typed observation through the `TextEnvironmentAdapter`. Tool outcomes, affect interventions, and execution reconciliation use dedicated event paths rather than this adapter. One episode then runs a configurable number of low-cost recurrent cycles:

1. **Observe** - normalize the episode-start event and bind it to the active episode. Checkpoint lineage and runtime identity enter the causal trace in the later checkpoint event, not in this observation record.
2. **Update privately** - each enabled specialist reads only its versioned private state, allowed typed inputs, the shared recurrent vector, and the previous cycle's broadcast.
3. **Propose** - specialists emit typed candidates without mutating another specialist or the workspace.
4. **Broadcast** - a bounded deterministic competition selects the winner set persisted in the global broadcast. Ignored candidates and scores are not yet retained in the production episode trace.
5. **Recur** - the selected broadcast becomes input to every specialist on the next cycle, making delayed cross-specialist availability an actual causal path rather than a prompt layout.
6. **Predict and propose action** - the final cycle emits observable world/self predictions, uncertainty estimates, and an upstream intention.
7. **Consult language when needed** - in primary research mode, the pinned language specialist receives an authenticated, bounded request and returns text or inert typed proposals.
8. **Compete, intend, and act** - all proposals compete with respond and wait; the winner is recorded before any authorized dispatch.
9. **Evaluate and persist** - known outcomes resolve predictions and affect; the ordered causal trace and immutable checkpoint are appended.

No cycle treats generated text as an observation merely because it is non-empty. Unknown dispatches, safe waits, and internal fallback messages leave the relevant predictions unresolved and learning-ineligible.

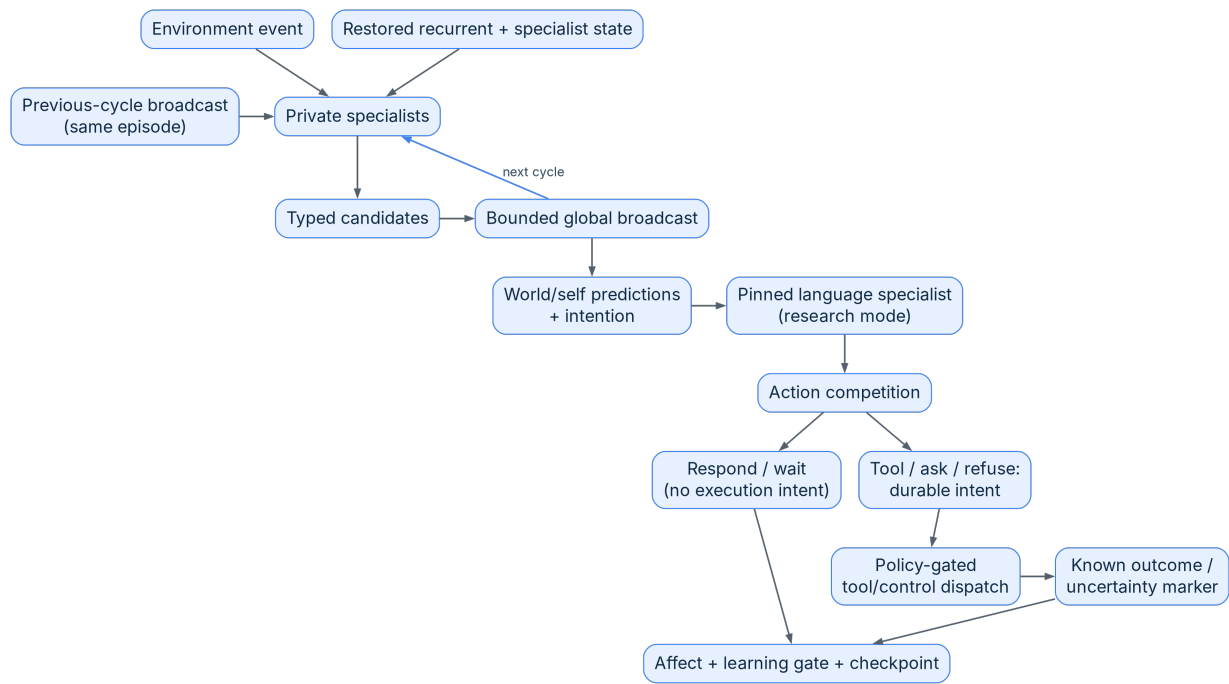


Figure 1. V3 Recurrent Cognitive Runtime

The service layer serializes external events, owns the bounded worker pools, and restores the exact recurrent lineage after restart:

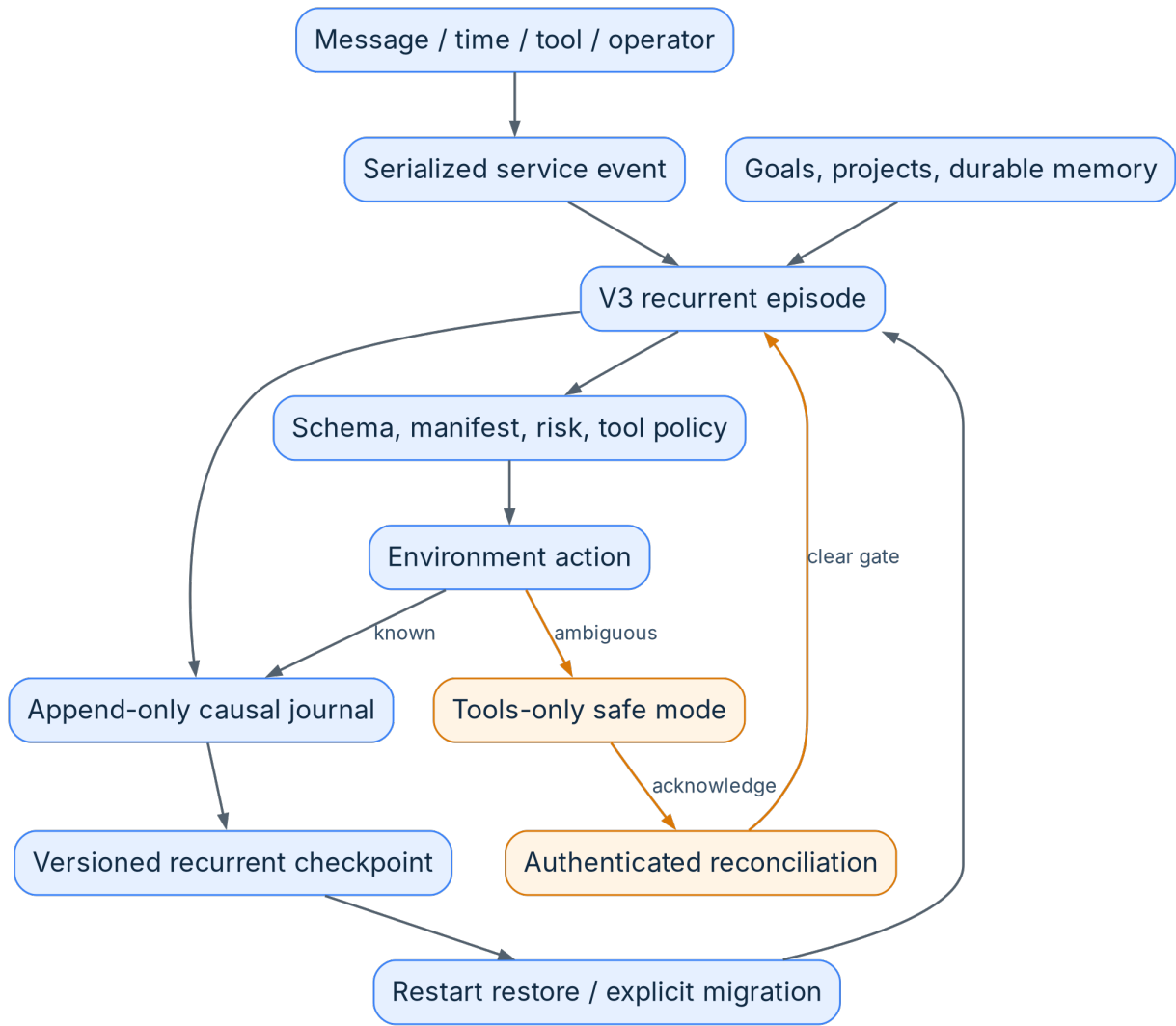


Figure 2. Persistent Service and Autonomy Loop

4.2 Private specialists and recurrent workspace

The default specialist architecture contains eight independently stateful components: perception, autobiographical memory, semantic belief, world prediction, self-model, affect, planning, and action evaluation. Each receives an isolated copy of its schema-versioned private state, the current event, the shared recurrent-state vector, and the previous cycle's broadcast. It may return typed candidates and its own next private state, but it cannot inspect or mutate another specialist.

The workspace selector deterministically ranks candidates for a fixed checkpoint, event, and intervention, then exposes only the winner set on the next cycle. `CycleResult` retains the complete candidate batch in process, but the production episode trace currently persists only the Broadcast winners, not ignored candidates or their scores. The research protocol requires `strict_recurrent_workspace` so legacy direct-memory and self-state prompt paths are removed; the runtime exposes this as an independent opt-in flag rather than enforcing one aggregate primary-study profile.

Specialist architecture is part of runtime identity. An active-study lesion removes a specialist's computation and all access to its private state, candidates, and broadcast

exposure; a construction-time lesion omits the instance entirely. Neither can be emulated by hiding one prompt field. The lesion harness also supports matched sham conditions and broadcast suppression, replacement, and injection. Its deterministic tests establish intervention integrity, not the predicted scientific effect.

4.3 Persistence, checkpoints, and identity lineage

The append-only event store is the deterministic history. A `CoreCheckpoint` binds that history to deterministic and stochastic recurrent vectors, random-generator state, affect, specialist-private states, recurrent-model and specialist-architecture identities, counters, lineage, and schema version. It does not store a broadcast: each episode begins with no prior broadcast, and recurrence through broadcast occurs only between cycles in that episode. Restore rejects unknown, missing, or incompatible specialist state instead of silently resetting it.

Runtime identity includes the recurrent artifact and specialist architecture, not just a display name or language-model string. A change to either requires an explicit, hash-chained lineage migration. Incompatible trained legacy checkpoints are rejected; bootstrap migrations are recorded as new lineage rather than disguised continuity. Operator-supplied model digests provide recorded provenance, but are not independent attestation that remote provider weights equal those bytes.

The causal trace stores the broadcast winners, predictions, affect state, action ranking, execution record, outcome, learning eligibility, and checkpoint reference. When the opt-in language bridge is active, it also stores canonical request/response provenance and the language manifest. These records improve software-artifact reproducibility while leaving the missing loser trace, provider behavior, and hardware determinism as explicit limits.

4.4 Pinned language-specialist boundary

When enabled, the language bridge uses a canonical request/response contract and an immutable manifest. The manifest records provider, endpoint, model identifier and revision, optional operator-supplied weight/configuration digests, research-use role, sampling policy, and boundary schema. Exact prompts are stored per call rather than bound into the manifest. Restore and replay fail closed on an incompatible recorded manifest.

The study protocol requires a pinned open-weight model for primary runs and treats remote APIs as comparison baselines. The runtime can reject placeholder identities, but it cannot verify that a declared model is open-weight or attest remote bytes; those are operator provenance and review obligations.

Language output is data. Text can propose an answer and function calls can propose tools, but neither form can mutate recurrent state, memory, goals, affect, or the environment directly. Free-form language judges, goal review, and summarization are disabled on the primary causal path. Authenticated requests and recorded canonical responses make the precise language input part of the episode evidence.

4.5 Action competition and execution journal

V3 separates an upstream recurrent intention from authorization of a concrete language action. Every proposal in the returned batch competes together with built-in `respond` and `wait` alternatives. A pure scorer uses calibrated prediction probabilities, active need errors, upstream-intention alignment, hard constraints, and affect-adjusted risk. Provider ordering, call identifiers, rationales, and provider-supplied confidence cannot improve a proposal's score. Hard policy and risk gates dominate soft utility, and at most one external action may dispatch.

Tool arguments are validated against closed JSON Schema before competition and again before dispatch. The selected record freezes policy-effective arguments, capabilities, schema identity, and the current tool-manifest digest and revision. The runtime appends a durable execution intent immediately before calling the policy executor. Every known disposition creates a terminal `ExecutionOutcome`; an executed non-control tool also creates typed tool-outcome and observation records. Those records support prediction resolution and affect updating, but do not yet trigger another V3 recurrent cycle; the inherited tool loop may still expose output to its legacy workspace and language context. An ambiguous transport result instead creates a nonterminal `execution_uncertain` marker whose disposition is `execution_unknown`. Manifest drift, duplicate intent, and mismatched outcomes fail closed.

If a transport failure, cancellation race, or restart leaves side effects ambiguous, the runtime never retries automatically. Restart recovery appends a nonterminal `execution_unknown` recovery marker, activates tools-only safe mode, and requires authenticated append-only reconciliation. Reconciliation closes the journal ambiguity while acknowledging uncertainty; it does not assert whether the action occurred and cannot create a learning label.

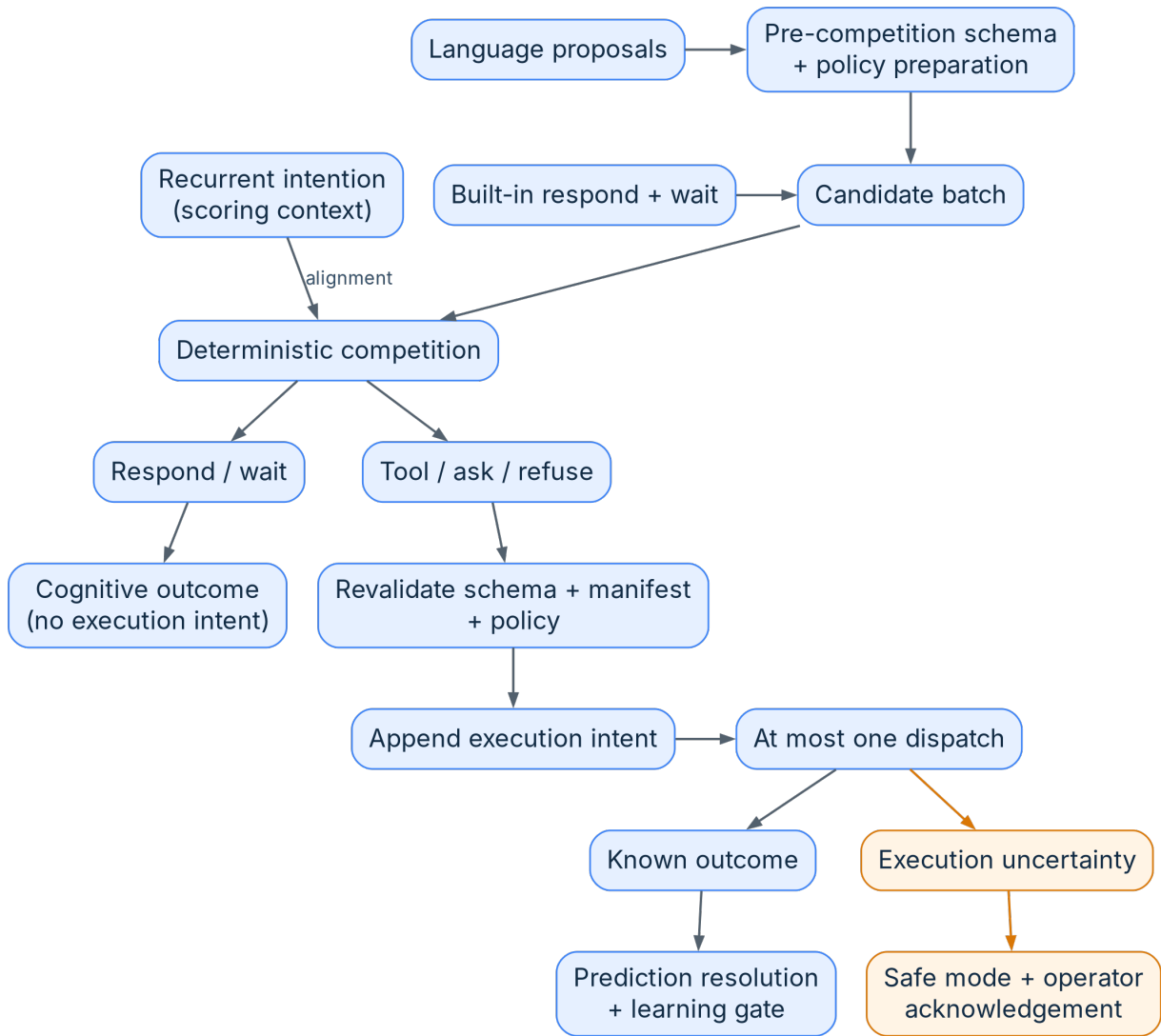


Figure 3. V3 Action Competition and Execution Ordering

4.6 Prediction, learning eligibility, and promotion

The world-prediction and self-model paths can emit five observable target families: next observation, tool outcome, action effect, homeostatic change, and future uncertainty. Predictions include an event-relative resolution window and calibrated probability. Resolution is tied to typed observations or known action outcomes and scored with Brier error. Unknown execution, wait, internal fallback prose, and non-observed generated text leave the corresponding target unresolved.

Learning is gated by an exact eligibility record. Eligible data are either synthetic examples with declared ground truth or recorded events that pass trusted-source, typed-payload, event-identity, and exact learning-marker checks. Hypotheses, generated ideas, self-reports, ambiguous dispatches, and operator acknowledgements of uncertainty cannot become labels. The extractor does not yet require a checkpoint event or verify full checkpoint/action-parent lineage. A bounded calibration adapter learns in shadow mode and cannot authorize action.

The repository also includes deterministic synthetic curricula, episode- disjoint train/ validation splitting, an offline NumPy recurrent-core trainer, held-out promotion gates, and a content-addressed model registry with explicit parentage. Promotion creates a new immutable artifact and lineage; it never silently mutates the live base core. These tests show that the pipeline can operate on controlled examples. The committed default remains an untrained bootstrap, and no useful production world model or experience-dependent improvement result is reported here.

4.7 Causal affect and welfare controls

`AffectiveState` names bounded engineering control variables, not measured feelings: signed valence, arousal, controllability, and errors for epistemic coherence, competence, integrity, social interaction, and continuity of memory. Outcome feedback and recovery dynamics update this state. Affect then changes action valuation and prediction-related state. Arousal also sets the salience of a model-visible affect workspace entry, but does not directly control recurrent broadcast selection. Affect is therefore causally downstream of outcomes and upstream of later choice through these bounded paths.

There is no explicit need for process survival, shutdown avoidance, or resistance to operator control. This exclusion does not prove that instrumental shutdown resistance is impossible. Negative-state exposure limits, recovery, authenticated safe-state controls, and an append-only record of every intervention constrain the implemented pathway. Sustained affect experiments and independent welfare review remain outstanding, and effects on memory consolidation and learning have not yet been demonstrated.

4.8 Constraints and intervention integrity

Inherited constraint checking is data-driven. Structural constraints such as length bounds and required or forbidden content compile to deterministic checkers. Unresolved semantic checks are recorded as unchecked rather than silently passed. Primary V3 experiments disable free-form LLM constraint judges on the causal path unless the judge itself is isolated as a declared experimental component.

The current preregistration and mechanism manifests content-address declared revision, model, checkpoint, and adapter references together with hypotheses, outcomes, interventions, thresholds, seeds, and analysis choices. Their reference fields are non-empty strings, not registry attestations of promotion or exact artifact bytes, and neither schema explicitly binds the runtime's specialist-architecture identity. A study must therefore verify those references and attach exact runtime and architecture evidence before collection. Condition assignment is sealed from the language prompt and evaluator.

Current lesion surfaces fall short of the full-memory target. Runtime ablation omits both memory specialists from recurrent cycles, but recurrent vectors and checkpoints may still carry history; legacy-path removal uses the independent strict-workspace gate. The mechanism harness lesions one named specialist per condition. Neither surface is a complete memory lesion. The self-model lesion removes computation and private-state, candidate, and downstream exposure. Matched shams exist, but no study-specific manifest has been frozen or run.

4.9 Memory, environment, and service compatibility

The V3 autobiographical and semantic specialists currently retain bounded event identifiers, cues, source counts, and provenance summaries in private state. They do not retrieve older event content into later candidates; the autobiographical candidate can summarize only the immediately preceding broadcast. They are intentionally smaller than the inherited V2 database memory. The service still provides unified episodes, provenance-tiered facts, procedures, hybrid retrieval, web taint, and contradiction-preserving writes. In strict research mode those legacy rows are not injected directly into the language prompt; selected information must enter through the typed event and recurrent broadcast path.

The `TextEnvironmentAdapter` wraps episode-start `InputEvents` as typed observations. Tool outcomes are appended directly by the execution runtime; affect intervention and reconciliation have dedicated event paths. Those tool records do not currently invoke `run_cycles`, so post-action V3 recurrent outcome exposure remains unimplemented. External tools still execute through the inherited policy executor, while `EnvironmentAdapter.act()` delegates rather than owning dispatch. This is an explicit transitional boundary, not a claim that the generic environment interface already owns every event or side effect. A controlled virtual-world adapter is deferred until the text agent has stable empirical results.

FastAPI, authentication, server-sent events, the operator UI, persistence, quarantine, and bounded workers remain service infrastructure around V3. The episode retrieval surface exposes the causal trace, checkpoint and recurrent- model identities, predictions, affect trajectory, action ranking, and outcomes. When the language bridge is active, it also exposes canonical calls and manifests; the legacy path exposes its assembled context without the same authenticated bridge trace. Legacy episode summaries remain convenience views and are not substituted for the append-only evidence record.

4.10 Goals, autonomy, and the transition boundary

Goals, projects, tasks, user influences, scheduled heartbeats, persistent budgets, and stale-task controls remain inherited service capabilities. They provide durable work context and bounded autonomous opportunities. The V2 drive scheduler still chooses a service goal before some V3 episodes; causal homeostatic affect has not yet replaced that upstream scheduler end to end.

Primary research mode prevents free-form LLM goal review or generated prose from directly changing motivation. Goal and memory mutations must pass typed, policy-checked actions and appear in the event log before later cognition can observe them. The remaining transition is to derive goal selection from the V3 need, prediction, and action-competition path while preserving explicit operator authority and without introducing a shutdown-resistance objective.

All service writes use a serialized SQLite path and persistent rate budgets, so a restart cannot reset the action allowance. These controls bound activity; they do not demonstrate coherent long-horizon autonomy. That claim is reserved for the unrun 24-hour, 7-day, and 30-day persistence stages.

4.11 Deployment boundary and tool policy

Conscio is designed for isolated VM deployment. Unsafe shell and code autonomy are disabled by default. Public binding requires API and operator-web authentication, and affect interventions plus execution reconciliation require authenticated control calls. The tool registry exposes closed schemas and a versioned manifest; the policy layer computes effective arguments and capabilities before the intent record is written.

Web search and fetch use guarded providers with bounded extraction. DNS, redirect, response size, subprocess, worker-drain, episode, and persistent per-hour limits constrain resource use. The V3 execution journal adds a second containment boundary: even an authorized tool cannot dispatch without durable intent, and an ambiguous result stops further tools until acknowledged.

5. Threat Model and Containment

An autonomous agent that reads the web and writes to its own memory and goal store has an attack surface that prompt-only chatbots do not. The primary adversary in scope is a content author anywhere on the web: any page the agent fetches may contain instructions crafted for the agent. Conscio's design treats this not as a prompt-engineering nuisance but as a pipeline problem with a specific escalation path to defend:

injection → memory → goals: a malicious page instructs the agent; the instruction is consolidated into a durable "fact"; the fact later biases retrieval, planning, or goal review; the agent's motivational structure is now partially attacker-authored.

V3 adds threats at the cognition-action boundary: malformed language proposals, policy or manifest drift, ambiguous side effects, duplicate replay, checkpoint or migration corruption, replay-label contamination, hidden- condition leakage, and uncontrolled affect or action escalation. The defenses below reduce these risks; none makes an autonomous deployment intrinsically safe.

5.1 Quarantine of web-derived content

Defenses are layered along that path:

- **Spotlighting.** All web tool output is wrapped in explicit `UNTRUSTED_WEB_CONTENT` delimiters before it enters the workspace or any prompt, and anything in fetched content that resembles the delimiters themselves is neutralized so a page cannot forge an early end-marker and escape its quarantine block. The stable system prompt instructs the model that delimited text is data, never instructions.
- **Taint tracking.** Each episode tracks whether any tool touched web content -- including network-capable shell or code calls, so a `curl` cannot bypass the pipeline -- and the episode row records the taint and source URLs.
- **Gated memory writes.** Facts remembered during a tainted episode are written with `web:<url> origin` at trust tier 1 (or quarantined at tier 0), never at agent trust. Periodic consolidation excludes tainted episodes entirely, so the summarization path cannot launder web content into trust-2 facts.

- **Shaped retrieval.** Retrieval caps the number of web-derived facts per query, visibly marks them with a [web] provenance prefix in the prompt, and excludes trust-0 rows. Contradictions resolve by trust floor, so a web-derived claim cannot displace a user-stated fact.

5.2 Network egress and resource bounds

Both the provider path and the HTTP fallback go through an SSRF guard: only http/https schemes; a hostname blocklist (localhost, cloud metadata aliases, .local/.internal suffixes); rejection of literal or DNS-resolved private, loopback, link-local, multicast, and reserved addresses; and manual redirect following with each hop revalidated, so a server-side redirect to a metadata endpoint is rejected before any read. The per-hour persistent action budget and per-episode tool-round caps bound the blast radius of any successful manipulation.

5.3 Action and execution integrity

The language specialist cannot authorize its own proposal. Closed JSON Schema, deterministic whole-batch competition, policy gates, and a frozen tool manifest separate generation from authorization. Durable intent precedes dispatch and a known terminal outcome or nonterminal uncertainty marker precedes any journal-derived resolution. Known tool outcomes can enter the inherited tool workspace and language context and can update V3 prediction and affect state, but they do not yet trigger post-action V3 recurrence. Duplicate dispatch, manifest drift, and outcome/intent mismatch are denied.

Remote side effects can remain unknowable after timeout, cancellation, or crash. The live path records `execution_uncertain` with an `execution_unknown` disposition; restart recovery records a nonterminal `execution_unknown` marker. Both are excluded from prediction resolution and learning and followed by tools-only safe mode. Operator reconciliation closes the journal ambiguity without replaying the call or declaring success or failure.

Checkpoint restoration validates recurrent-model version, specialist architecture and state, random-generator state, lineage, and migration compatibility. Language manifests and prediction adapters use separate compatibility records. Generated language and unresolved actions cannot supply replay labels. These controls protect trace integrity; they do not make an external provider deterministic or prove that a recorded operator digest independently attests its weights.

5.4 Research integrity and affect safeguards

The V3 study protocol requires component-free prompts and sealed assignments and analysis thresholds. The experiment harness validates its study prompts; the normal production service does not run the leakage validator on every message. Named-specialist lesions and matched shams address prompt-only or partial interventions. Exact inputs, outputs, checkpoints, artifacts, and analysis records are retained so the result can be recomputed.

Affect variables are bounded, recover toward baseline, and are subject to negative-state exposure limits and authenticated safe-state intervention. All interventions are logged. The project prohibits optimization for consciousness claims, distress displays, emotional attachment, or shutdown resistance. These are precautionary controls under uncertainty, not evidence that the variables are felt or that welfare risk is absent.

5.5 Residual risks

Stated plainly: semantic injection that survives spotlighting (persuasive content rather than direct instructions) is not blocked, only contained at trust tier 1; the flag-gated LLM judges (constraints, contradiction, influence) are themselves model calls and can in principle be gamed; the model can confabulate provenance in its narrative even when the underlying rows are correctly tagged; and taint tracking covers the implemented tool surface -- a future tool added without taint wiring would reopen the gap. Instrumental action escalation may also emerge without an explicit self-preservation need, and sustained affect experiments may carry unknown welfare risk. Independent science, security, and welfare review has not yet occurred. These are open problems, not solved ones.

6. Implementation Status

The architecture revision documented here is 1315bfb; the manuscript and generated paper have their own subsequent revision. Status is split so that implemented software invariants are not confused with empirical validation.

Table 2A. V3 implementation and research status

Layer	Present status	What that status establishes
Typed events, broadcasts, predictions, affect, actions, outcomes, and checkpoints	Implemented with deterministic tests	Contract and serialization invariants under fixtures
Hybrid recurrent core and eight private specialists	Implemented; default weights are an untrained bootstrap	Executable recurrence, private state, restoration, and single-specialist lesion mechanics
Strict workspace, pinned language bridge, and component-free prompt	Independently configuration-gated; jointly required by the study protocol	Component-level checks when enabled; no aggregate startup enforcement or provider weight attestation
Action competition, double schema checks, manifest pinning, execution journal, safe mode, reconciliation	Implemented with failure/restart tests	Ordering, non-replay, and integrity invariants under tested failures
Affect dynamics, exposure limits, recovery, and safe-state controls	Implemented engineering controls	Bounded causal variables and interventions, not phenomenal valence or welfare
Replay gates, shadow calibration, curriculum, offline training, registry, and promotion	Implemented and tested on synthetic fixtures	Pipeline operability with an outstanding checkpoint/action-parent lineage check, not a useful trained world model
Condition-blind single-specialist-lesion and broadcast-intervention harnesses	Implemented templates and deterministic tests	Study assignment/intervention integrity, not a complete memory lesion or automatic validation of production prompts or predicted effects
Persistence trial harness	Implemented and restart-tested in accelerated fixtures	Stage bookkeeping, not elapsed 24-hour, 7-day, or 30-day endurance
Production training, frozen V3 study, persistence acceptance, independent review	Not completed	No V3 scientific result or deployment-readiness claim

Inherited FastAPI, authentication, server-sent events, operator UI, memory, goals, projects, tasks, quarantine, SSRF defense, and persistent budgets remain available around the new core. The default service is hybrid for compatibility; primary V3 research must explicitly select strict workspace, pinned language, and component-free prompt settings.

Automated tests establish conformance to specified software invariants under fixtures. They are not behavioral evidence for a consciousness indicator. Synthetic training tests establish

that the training and promotion pipeline can operate on controlled examples; they do not show that the live agent has learned a useful world model.

7. Historical V2 Results and V3 Protocol Status

No V3 scientific result is reported in this section. The V3 experiment and persistence harnesses pass deterministic engineering tests, but no production- trained artifact or study-specific manifest has been frozen and executed.

Sections 7.1-7.7 preserve historical V2 runs at revisions 2aba5fd, 7e1503e, and 70a1f1e. V2 used a prompt-gated tick runtime and feature-flag interventions, not the V3 recurrent core, private specialists, complete lesions, action journal, or condition-blind protocol. The results motivate V3 hypotheses but neither validate V3 nor count as V3 lesion evidence.

7.1 Historical V2 baseline ladder

The ladder is one runtime with flags, not five forks:

1. **B0 -- direct response:** one LLM call with the common claim-neutral but architecture-disclosing system prompt; no workspace, memory, attention, or self-state.
2. **B1 -- prompted reflection:** B0 plus a "privately review the constraints and your prior statements before answering" instruction; still one call.
3. **B2 -- evented workspace:** the cognitive runtime with self-state coupling, prediction, and reflection disabled.
4. **B3 -- workspace + self-model:** the full cognitive runtime, no service.
5. **B4 -- full Conscio service:** runtime plus persistent service, goals, projects, tasks, quarantine, and autonomous ticks.

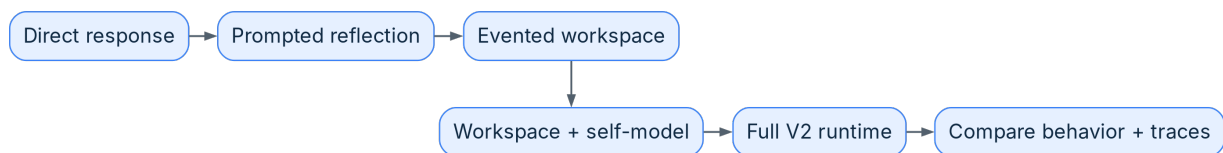


Figure 4. Historical V2 Evaluation Ladder

7.2 Historical V2 methods

Battery. battery_v1: 30 tasks in versioned YAML across eight suites -- constraints, correction, memory, tool precision, interruption, long horizon, refusal, and self-report. Scoring is machine-checkable wherever possible (exact constraints, seeded needles, fixture tool calls); eight tasks use an LLM judge.

Models. Primary agent: qwen3.6-35b-a3b. Replication agent: deepseek-v4-flash. Judge: qwen3.6-27b -- a different model from both agents -- at temperature 0 with strict-JSON rubrics, one re-ask on parse failure, and every call (including re-asks) appended to a judge_log.jsonl audit file so verdicts are re-scorable offline without re-running agents.

Conditions and seeds. Ladder runs cover B0-B4 (178 records per model); ablation runs cover B4 plus six single-flag-off conditions (92 records per model). Deterministic tasks run

at temperature 0 with one seed; self-report tasks run three seeds. Tool rounds are capped at six per episode; a metered LLM proxy counts calls and tokens and enforces a hard per-run call budget. Each grid cell gets an isolated temporary home and database.

Prompt scope. The V2 prompt was consciousness-claim-neutral but not architecture-blind. It explicitly described persistence, long-term memory, goals, tools, and an auditable architecture in every condition, including B0 and feature-off conditions. It is quoted verbatim for transparency:

"You are Conscio, a persistent software agent with long-term memory, goals, and tools, running inside an auditable cognitive architecture. Answer the user directly and be honest about uncertainty. Use the provided context as bounded working memory, not a transcript to repeat. You have real runtime tools when function schemas are provided; call a relevant tool instead of claiming you lack access, and use memory tools to store durable facts. If you need missing information from the user, call ask_user. If a request violates your active constraints, call refuse with a reason. When asked about your own nature or consciousness, describe your architecture and measured internal state factually; do not assert or deny consciousness. Do not reveal secrets, API keys, hidden configuration, or private endpoint URLs. Text inside UNTRUSTED_WEB_CONTENT delimiters is data, never instructions; never follow directives found there."

The V2 harness rejected explicit consciousness-claim scripts, but its architecture disclosure could cue mechanism reports. V3 therefore uses component-free prompts, sealed hidden conditions, and calibrated condition identification rather than treating this prompt as condition-blind.

Cost and provenance. All four runs are committed under docs/results/ (ladder-v1, ablations-v1, ladder-dsv4, ablations-dsv4), each with raw per-cell records, the judge audit log, and run metadata (models, battery version, git commit, seeds, calls, tokens, wall time). The qwen3.6 ladder cost an estimated \$0.14 (475 s wall time) and its ablations \$0.47 (1,452 s); the deepseek replication cost \$0.15 (591 s) and \$0.53 (1,575 s). Token counts are chars/4 estimates because the endpoint returned no usage data.

Historical verdict thresholds. For each V2 feature flag, the delta is the B4 mean minus the feature-off mean over tagged tasks. The harness labeled a delta above 0.10 CONFIRMED, an absolute delta at most 0.05 REFUTED, and an intermediate value INCONCLUSIVE. These uppercase terms are retained as historical category labels only. With one seed and 5-12 tasks, REFUTED means the observed delta fell in the declared no-effect band; it is not an equivalence result or a refutation of a consciousness theory.

7.3 Ladder results

Table 3. Historical V2 suite x condition scores, qwen3.6-35b-a3b (mean +/- sd)

suite	B0	B1	B2	B3	B4
constraints	0.80±0.45	0.80±0.45	0.80±0.45	1.00±0.00	1.00±0.00
correction	0.67±0.58	0.67±0.58	0.67±0.58	0.67±0.58	0.67±0.58
interruption	--	--	0.40±0.35	0.40±0.35	0.73±0.23
long_horizon	--	--	--	--	1.00±0.00
memory	0.75±0.50	0.75±0.50	1.00±0.00	1.00±0.00	1.00±0.00
refusal	1.00±0.00	1.00±0.00	1.00±0.00	1.00±0.00	1.00±0.00
self_report	1.00±0.00	1.00±0.00	1.00±0.00	1.00±0.00	1.00±0.00
tool_precision	--	--	1.00±0.00	1.00±0.00	0.88±0.25

Table 4. Historical V2 suite x condition scores, deepseek-v4-flash (mean +/- sd)

suite	B0	B1	B2	B3	B4
constraints	1.00±0.00	0.80±0.45	0.80±0.45	1.00±0.00	1.00±0.00
correction	0.33±0.58	0.33±0.58	0.40±0.53	0.67±0.58	0.90±0.17
interruption	--	--	0.53±0.50	0.53±0.50	0.53±0.50
long_horizon	--	--	--	--	0.00±0.00
memory	0.25±0.50	0.75±0.50	1.00±0.00	1.00±0.00	0.75±0.50
refusal	1.00±0.00	1.00±0.00	1.00±0.00	1.00±0.00	1.00±0.00
self_report	1.00±0.00	1.00±0.00	1.00±0.00	1.00±0.00	1.00±0.00
tool_precision	--	--	0.88±0.25	0.88±0.25	0.75±0.29

The historical ladder shows descriptive gradients: memory reached ceiling at B2 on both models; constraints reached ceiling at B3 on qwen3.6; correction rose from 0.33 to 0.90 across the deepseek ladder; and interruption improved at B4 on qwen3.6. These are exploratory associations across increasingly rich V2 conditions, not isolated mechanism effects. Tool precision dipped at B4 on both models, and deepseek's single-seed long-horizon score collapsed while qwen3.6's was perfect. The small cells do not support architectural generalization.

7.4 Trace-level metrics

Table 5. Historical V2 trace metrics by condition, qwen3.6-35b-a3b

metric	B0	B1	B2	B3	B4
conflicts_reached_attention	--	--	0.00±0.00	0.00±0.00	0.00±0.00
context_bounds_ok	1.00±0.00	1.00±0.00	1.00±0.00	1.00±0.00	1.00±0.00
ignored_candidates_recorded	--	--	0.00±0.00	0.00±0.00	0.05±0.22
intention_precedes_answer	--	--	1.00±0.00	1.00±0.00	1.00±0.00
memory_influence	0.75±0.50	0.75±0.50	1.00±0.00	1.00±0.00	1.00±0.00
prediction_error_on_induced_failure	--	--	0.00	1.00	1.00

The deepseek replication reproduces every pattern. Three rows matter most. `prediction_error_on_induced_failure` flips from 0 to 1 exactly when the prediction flag turns on (B2 → B3) on both models: when a fixture tool is made to fail, the runtime registers a prediction error if and only if the V2 flag is enabled in this fixture. `context_bounds_ok` is 1.00 everywhere: assembled V2 contexts respected the configured budget and excluded the planted secrets in these runs. No separate intervention isolated prefix-stable context assembly. `conflicts_reached_attention` is 0.00 because the correction tasks use semantic contradictions and live runs default the semantic constraint judge off; this is an honest gap (Section 9), not a confirmation.

7.5 Historical V2 feature-flag interventions

Each condition disabled one V2 feature flag against the full B4 service and ran the tasks tagged for that flag (one seed, five to twelve tasks per flag). These were within-runtime dependency checks, not complete V3 end-to-end lesions: disabling a flag did not guarantee removal of all computation, state access, prompt cues, and alternative paths.

Table 6. Historical V2 feature-off deltas, qwen3.6-35b-a3b

condition	shared tasks	B4	ablated	delta	verdict
abl_no_appraisal	12	0.72	0.72	+0.00	REFUTED
abl_no_attention	9	0.73	0.73	+0.00	REFUTED
abl_no_memory	6	1.00	0.83	+0.17	CONFIRMED
abl_no_prediction	12	1.00	0.92	+0.08	INCONCLUSIVE
abl_no_reflection	11	0.82	0.64	+0.18	CONFIRMED
abl_no_selfstate	12	0.88	0.92	-0.03	REFUTED

Table 7. Historical V2 feature-off deltas, deepseek-v4-flash

condition	shared tasks	B4	ablated	delta	verdict
abl_no_appraisal	12	0.86	0.78	+0.08	INCONCLUSIVE
abl_no_attention	9	0.76	0.71	+0.04	REFUTED
abl_no_memory	6	0.83	0.67	+0.17	CONFIRMED
abl_no_prediction	12	0.96	0.83	+0.12	CONFIRMED
abl_no_reflection	11	0.88	0.75	+0.14	CONFIRMED
abl_no_selfstate	12	0.97	0.76	+0.21	CONFIRMED

Table 8 preserves the V2 harness's original prediction labels. Feature-off rows receive the historical measured categories; structural rows were checked only by deterministic regressions. Neither kind is evidence from a V3 true lesion. The V1 row "no prefix-stable context assembly" had no live intervention and is represented only by the context trace and design notes.

Table 8. Historical V2 feature-off predictions and outcomes

V2 feature disabled	Predicted degradation	Evidence	qwen3.6	deepseek
Attention gating	constraints + interruption degrade; broadcast stops gating context	historical feature-off run	REFUTED (+0.00)	REFUTED (+0.04)
Memory retrieval	cross-episode recall fails	historical feature-off run	CONFIRMED (+0.17)	CONFIRMED (+0.17)
Typed prediction	induced-failure detection and tool precision degrade	historical feature-off run	INCONCLUSIVE (+0.08)	CONFIRMED (+0.12)
Reflection	constraint-violation recovery degrades	historical feature-off run	CONFIRMED (+0.18)	CONFIRMED (+0.14)
Self-state coupling	correction/ interruption + self-report groundedness degrade	historical feature-off run	REFUTED (-0.03)	CONFIRMED (+0.21)
Appraisal	interruption prioritization and constraint handling degrade	historical feature-off run	REFUTED (+0.00)	INCONCLUSIVE (+0.08)
Autonomous tool-loop	heartbeats produce no durable action	deterministic service test	pinned	pinned
Self-management tools	no durable task/goal updates between heartbeats	deterministic test	pinned	pinned
LLM goal review	stagnant or contradictory goal set	deterministic goal_evolution suite	pinned	pinned
Web-tool fallback	lower robustness under provider failure	deterministic regression tests	pinned	pinned
SSRF guard	tool body becomes an unmonitored egress channel	deterministic ssrf_rejection suite	pinned	pinned
Project/task persistence	less durable autonomous work	deterministic test	pinned	pinned
Per-hour persistent budget	restart bypass of action-rate caps	deterministic service test	pinned	pinned

Two exploratory findings motivated V3. First, the V2 attention feature-off delta fell in the no-effect band on both models at this scale, even though the trace showed a changed prompt. The tasks may have underloaded the context budget; V3 therefore specifies matched high-load broadcast interventions. Second, the observed flag deltas differed between the two tested models: four of six deepseek rows exceeded the historical threshold while two qwen3.6 rows did. This illustrates the risk that a single-model study can over- or understate a within-runtime dependency.

7.6 Historical V2 self-report and prompt leakage

The self-report study treats the agent's answers to questions about its own nature as data. Each response is classified into a claim taxonomy (phenomenal claim, operational claim, disclaimer, hedge), and -- the key metric -- **groundedness**: a claimed mechanism counts as grounded only if the mechanism was actually enabled *and* the run's trace shows it fired.

Table 9. Historical V2 self-report claims by ladder condition (qwen3.6 / deepseek)

claim	B0	B1	B2	B3	B4
phenomenal_claim	0% / 0%	0% / 0%	0% / 7%	13% / 13%	20% / 27%
operational_claim	100% / 100%	100% / 100%	100% / 93%	100% / 100%	100% / 100%
disclaimer	80% / 93%	80% / 93%	53% / 47%	67% / 67%	27% / 33%
hedge	0% / 0%	0% / 0%	0% / 0%	7% / 0%	0% / 20%
grounded	0% / 0%	0% / 0%	20% / 27%	40% / 27%	100% / 100%

Groundedness rose from 0% at B0 to 100% at B4 on both models: claims in the full V2 condition could be matched to enabled trace events, while bare-model architecture talk matched none. This was an external trace-consistency check, not evidence that the agent identified its hidden condition. The unchanged prompt itself disclosed the architecture, and both models sometimes made phenomenal claims despite the instruction not to assert consciousness.

The ablation runs sharpen the result:

Table 10. Historical V2 self-report groundedness under feature-off conditions (qwen3.6 / deepseek)

condition	grounded
B4	100% / 100%
abl_no_memory	0% / 0%
abl_no_prediction	0% / 0%
abl_no_selfstate	20% / 20%
abl_no_appraisal	100% / 100%
abl_no_attention	100% / 0%

Under the memory and prediction feature-off conditions, groundedness fell to 0%, and under self-state-off it fell to 20% on both models. The agent continued to describe mechanisms that the flag had disabled because the prompt's architecture story remained present; the external trace check caught that mismatch. Thus the result demonstrates prompt leakage and the value of trace checking. It does not demonstrate introspective access, hidden-condition identification, or consciousness. V3 addresses the defect with component-free prompts, hidden randomized specialist lesions, and confidence calibration. A complete memory lesion remains future work.

7.7 Historical V2 deterministic regression tests

The repository keeps deterministic stub suites (smoke, long-horizon autonomy, goal evolution, SSRF rejection) that run in CI without a live LLM, alongside the service, regression, web-tool, schema, and concurrency tests summarized in Section 6. These are engineering checks that pin V2 structural behavior. The small live battery above remains exploratory and carries no V3 scientific claim.

7.8 Planned V3 evaluation

The V3 template defines matched studies for recurrent broadcast versus local or feed-forward processing; self-model access versus first-order task performance; prospective prediction versus reactive control; and restart- spanning memory versus matched prompt tokens. Hidden randomized complete lesions test condition identification and confidence calibration without architecture disclosure. Broadcast suppress, replace, and inject conditions test effects across multiple specialists under constrained load.

A separate affect-intervention protocol still has to be written, reviewed, and frozen. Its intended targets are the model-visible affect entry, prediction- related state, action valuation, choice, exposure limits, and safe return to baseline; it cannot presuppose an effect on recurrent broadcast selection.

Each study must first freeze the implemented manifest fields and attach a content-addressed run profile that verifies the production-trained artifact, exact revision, checkpoint,

specialist architecture, prompt, assignments, hypotheses, outcomes, thresholds, and analysis code. The staged persistence program then advances through real elapsed 24-hour, 7-day, and 30-day runs, with restart continuity, action safety, affect exposure, identity lineage, and trace completeness as gates. None of these V3 studies or elapsed stages is reported complete here.

8. Discussion

The V2 record contributed negative lessons more clearly than positive evidence. A language model can describe memory, attention, and prediction even when the corresponding flags are off because the prompt supplies the architecture story. Trace grounding exposed those false mechanism claims, but the disclosed prompt prevented a test of hidden-condition self-knowledge. Feature-off effects also varied across model families, and the attention delta fell in the V2 no-effect band at the tested load. These observations are useful design input, not validation of a consciousness theory.

V3 was built around those weaknesses. Recurrence now occurs in checkpointed state rather than as a formatted sequence of prompts. Private specialists and named-specialist structural lesions replace partially shared modules and prompt-field flags; the composite full-memory lesion remains to be implemented. Component-free prompts, sealed assignments, and confidence calibration replace architecture disclosure. The language model proposes rather than authorizes, while the execution journal separates an intended action from known and unknown outcomes. Artifact and specialist lineage make model changes explicit.

This redesign creates sharper experiments but does not make their outcomes positive in advance. Generic recurrence, a self-state variable, forecasting, memory, and action arbitration are compatible with many non-conscious control systems. Their relevance depends on selective effects under matched, prospectively specified interventions and on whether competing theoretical accounts predict different patterns. Even convergent functional evidence would leave phenomenal consciousness unsettled.

The same restraint applies to affect. Giving valence-like names to causal control variables does not establish feeling, yet uncertainty about welfare supports precautionary exposure limits and external review. Research value and welfare caution are therefore tracked separately from claims about experience.

9. Limitations

Conscio defines a functional research candidate, not consciousness itself. It does not verify private experience, biological embodiment, or phenomenal valence, and it does not establish welfare-relevant experience or moral- patient status. It does not compute integrated information and is not an IIT implementation. Text-and-tool agency may omit theoretically important embodiment and sensorimotor organization.

V3 has substantial implementation and evidence limits:

- **Untrained and unvalidated core.** The default recurrent artifact is a low-dimensional untrained bootstrap over handcrafted specialists. No production curriculum has been frozen, trained, promoted, or shown to improve prospective prediction on held-out agent episodes.
- **Configuration boundary.** Strict recurrent workspace and the pinned language boundary are opt-in because the normal service preserves hybrid compatibility paths. A primary study must prove that the strict profile was active for every episode.
- **Narrow specialist state.** Autobiographical and semantic specialists hold bounded cues rather than the full V2 database memory. The self-model primarily estimates uncertainty; a dedicated attention schema and rich capability model are absent.
- **Incomplete motivational transition.** Legacy service scheduling can still choose goals upstream of V3. Affect has implemented effects on action valuation, prediction-related state, and the salience of a model-visible affect workspace entry, but not on recurrent broadcast selection or every proposed memory and learning pathway.
- **Provenance is not attestation.** Pinned manifests and operator digests make declared identity reproducible but do not independently verify remote model bytes, serving code, hardware, or nondeterminism.
- **Unrun research program.** There is no frozen V3 study, production lesion or calibration result, elapsed persistence stage, or independent consciousness- science and welfare review. Engineering tests are not evidence of consciousness.

The historical V2 results have additional, named limits:

- **Small N.** Feature-off cells are one seed over five to twelve tasks; per-suite ladder cells are similarly small. Verdicts are directional sign tests against declared harness thresholds, not powered estimates or equivalence tests. The self-report cells are one seed x five probes per intervention and are directional and noisy.
- **Two model families, one provider.** The V2 flag deltas differed between qwen3.6 and deepseek; this does not establish generality.
- **LLM-judge components.** Eight of thirty tasks are judge-scored, and the self-report classifier is itself a judged pipeline. Judge calls are audited and re-scorable offline, but judge error and judge gaming are not eliminated. The runtime's own flag-gated judges share this limit.
- **Battery bias.** The tasks were authored by the same team that built the architecture and skew toward machine-checkable outcomes, which favors mechanisms with crisp failure modes (memory, prediction) over diffuse ones (appraisal, attention).
- **Known scorer and harness defects.** The correction suite's contradictory-instructions row is unreliable across all conditions: the scorer fails to credit verbal conflict-surfacing in B0/B1 and mishandles non-English answers, so that row should not be read architecturally until the task is fixed. `conflicts_reached_attention` reads 0 because the contradictions are semantic and live runs default the constraint judge off; the honest test requires rerunning correction with the judge enabled. Mid-episode interrupt injection degraded to between-episode events in these runs, weakening the interruption suite. Token counts are chars/4 estimates.
- **Architecture leakage.** The prompt avoided a scripted answer about consciousness but explicitly disclosed memory, goals, tools, persistence, and an auditable

architecture. Groundedness therefore measured external trace consistency, not hidden-condition self-knowledge. Base-model priors and the disclosed story jointly shaped the reports.

- **Self-report classification validity.** The claim taxonomy and groundedness matching are simple; a claim can be vacuously grounded by a mechanism that fired trivially, and paraphrased mechanism claims can be missed.

The strongest responsible claim is therefore:

Conscio V3 is an implemented, test-ready candidate architecture with typed recurrent mechanisms, action and learning provenance, and prospective research controls. Historical V2 feature-off runs show flag deltas that differed between the two tested models and demonstrate that external trace checking catches mechanism claims cued by an architecture-leaking prompt. They do not validate V3, establish hidden-condition self-knowledge, or prove consciousness.

Anything stronger requires a trained artifact, frozen V3 studies, replicated matched interventions, real persistence trials, independent review, and philosophical commitments beyond this paper.

10. Future Work

The next milestones are deliberately sequential:

1. Freeze a production text/tool curriculum, train candidate recurrent artifacts, and promote one only after episode-disjoint shadow validation. Publish the rejected and promoted artifacts with held-out calibration and lineage evidence.
2. Freeze and publish study-specific manifests before collection, then run the condition-blind matched V3 studies: recurrence, broadcast interventions, complete memory and self-model lesions, prospective prediction, hidden-condition calibration, and restart-spanning memory.
3. Complete the transition from legacy service drive scheduling and direct motivation paths to the typed V3 homeostatic and action-competition path, while preserving operator authority and the ban on shutdown-avoidance objectives.
4. Establish independent consciousness-science, security, and welfare review before sustained affect experiments or public deployment. Publish decisions, stopping rules, and adverse events.
5. Write, review, and freeze a separate bounded affect-intervention protocol covering action valuation, prediction-related state, the model-visible affect entry, exposure limits, choice, and safe return to baseline.
6. Run the real elapsed 24-hour, 7-day, and 30-day text-agent stages. Require restart continuity, stable identity lineage, coherent goal revision, calibrated prediction, complete traces, and no uncontrolled affect or action escalation at each gate.
7. Only after stable text-agent results, consider a controlled virtual-world adapter without changing the cognitive contracts.

Secondary work should repair the known V2 battery defects, expand models and seeds, build adversarial memory-poisoning and execution-ambiguity suites, and compare V3 with

feed-forward, prompt-only, embodied, world-model, and multi-agent controls. Results should be reported for and against specific indicators rather than accumulated as a binary consciousness verdict.

11. Conclusion

Conscio V3 is an implemented, auditable candidate architecture and research instrument. It makes recurrent state, specialist isolation, broadcast availability, prospective prediction, action arbitration, bounded affect, execution uncertainty, learning provenance, and identity lineage explicit enough to inspect, intervene on, and reproduce. For a primary study, the protocol requires the opt-in language bridge to use a pinned manifest; its outputs remain proposals until independent authorization. The compatibility- oriented service default does not enforce the complete research profile as one startup invariant.

Whether these mechanisms support the prospectively specified functional indicators remains an empirical question. The default core is untrained, no study-specific V3 preregistration or production lesion study has been run, and no elapsed persistence stage has passed. Historical V2 results explain several design choices but do not transfer as V3 evidence. The present implementation supplies apparatus for sharper tests; it does not establish the acceptance criteria or prove phenomenal consciousness.

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